SQL QUERY IMPLE MENTATION FOR E- COMMERCE DATABASE

Project Title: Meesho Jewelry Marketplace Database

Objectives

1. Perform SELECT, WHERE, ORDER BY, and DISTINCT queries.
2. Search products based on price, category, and availability.
3. Retrieve customer and product information.
4. Apply filtering conditions.
5. Generate basic business reports.

1. Basic SQL Queries

1.1 SELECT Query

The SELECT statement is used to retrieve data from a table**.**

PRODUCT_ID	PRODUCT_NAME	DESCRIPTION	PRICE	IMAGE	SELLER_ID
1	Gold Plated Necklace	Elegant gold plated necklace for women	599.00	gold_necklace.jpg	101
2	Diamond Earrings	Stylish diamond earrings for women	799.00	diamond_earrings.jpg	102
3	Pearl Bracelet	Beautiful pearl bracelet for women	450.00	pearl_bracelet.jpg	103
4	Silver Anklet	Traditional silver anklet design	350.00	silver_anklet.jpg	104
5	Stone Ring	Trendy stone ring for women	299.00	stone_ring.jpg	105

Display selected product details

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE
  2  FROM PRODUCT;

PRODUCT_ID PRODUCT_NAME                    PRICE
---------- ------------------------------ ----------
         1 Gold Plated Necklace           599.00
         2 Diamond Earrings               799.00
         3 Pearl Bracelet                 450.00
         4 Silver Anklet                  350.00
         5 Stone Ring                     299.00
```

Find products with price less than 1000

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE PRICE < 1000;

PRODUCT_ID PRODUCT_NAME                    DESCRIPTION                                PRICE IMAGE                    SELLER_ID
---------- ------------------------------ ---------------------------------------- ---------- ------------------------ ---------
         1 Gold Plated Necklace           Elegant gold plated necklace for women      599.00 gold_necklace.jpg              101
         2 Diamond Earrings               Stylish diamond earrings for women          799.00 diamond_earrings.jpg           102
         3 Pearl Bracelet                 Beautiful pearl bracelet for women          450.00 pearl_bracelet.jpg             103
         4 Silver Anklet                  Traditional silver anklet design            350.00 silver_anklet.jpg              104
         5 Stone Ring                     Trendy stone ring for women                 299.00 stone_ring.jpg                 105
```

Find products with price greater than 100

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE PRICE > 100;

PRODUCT_ID PRODUCT_NAME                 DESCRIPTION                                PRICE IMAGE                 SELLER_ID
---------- ---------------------------- ---------------------------------------- ---------- --------------------- ---------
         1 Gold Plated Necklace         Elegant gold plated necklace for women      599.00 gold_necklace.jpg           101
         2 Diamond Earrings             Stylish diamond earrings for women          799.00 diamond_earrings.jpg        102
         3 Pearl Bracelet               Beautiful pearl bracelet for women          450.00 pearl_bracelet.jpg          103
         4 Silver Anklet                Traditional silver anklet design            350.00 silver_anklet.jpg           104
         5 Stone Ring                   Trendy stone ring for women                 299.00 stone_ring.jpg              105
```

1.3 ORDER BY

The ORDER BY clause is used to sort the query results.

Products from lowest to highest price

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE
  2  FROM PRODUCT
  3  ORDER BY PRICE ASC;

PRODUCT_ID PRODUCT_NAME                      PRICE
---------- ------------------------------ ----------
         5 Stone Ring                        299.00
         4 Silver Anklet                     350.00
         3 Pearl Bracelet                    450.00
         1 Gold Plated Necklace              599.00
         2 Diamond Earrings                  799.00
```

Products from highest to lowest price

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE
  2  FROM PRODUCT
  3  ORDER BY PRICE DESC;

PRODUCT_ID PRODUCT_NAME                      PRICE
---------- ------------------------------ ----------
         2 Diamond Earrings                  799.00
         1 Gold Plated Necklace              599.00
         3 Pearl Bracelet                    450.00
         4 Silver Anklet                     350.00
         5 Stone Ring                        299.00
```

1.4 DISTINCT

The DISTINCT keyword is used to display unique values.

Display unique categories

```
SQL> SELECT DISTINCT CATEGORY_NAME
  2  FROM CATEGORY;

CATEGORY_NAME
----------------------------------------
--------------------
Bangles
Bracelets
Earrings
Necklaces
Rings
```

2. Search Products

2.1 Search Products Based on Price

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE
  2  FROM PRODUCT
  3  WHERE PRICE BETWEEN 100 AND 400;

PRODUCT_ID PRODUCT_NAME                    PRICE
---------- ------------------------------ ----------
         4 Silver Anklet                  350.00
         5 Stone Ring                     299.00
```

2.2 Search Products Based on Category

Your current PRODUCT table does not contain a CATEGORY_ID column. Therefore, you can display the available jewelry categories using:

```
    SELECT CATEGORY_ID, CATEGORY_NAME
  2  FROM CATEGORY
  3  WHERE CATEGORY_ID = 2;

CATEGORY_ID CATEGORY_NAME
----------- ------------------------------
--------------------------------
          2 Necklaces
```

3. Retrieve Customer Information

Display all customers

```
SQL> SELECT * FROM CUSTOMER;

CUSTOMER_ID CUSTOMER_NAME            EMAIL                              PHONE_NUMBER    ADDRESS
----------- ------------------------ ---------------------------------- --------------- ---------------
          1 Priya                    priya@gmail.com                    9876543210      Chennai
          2 Anu                      anu@gmail.com                      9876543211      Madurai
          3 Divya                    divya@gmail.com                    9876543212      Coimbatore
          4 Kaviya                   kaviya@gmail.com                   9876543213      Salem
          5 Meena                    meena@gmail.com                    9876543214      Trichy
```

Search for a particular customer

```
SQL> SELECT *
  2  FROM CUSTOMER
  3  WHERE CUSTOMER_ID = 1;

CUSTOMER_ID CUSTOMER_NAME            EMAIL                              PHONE_NUMBER    ADDRESS
----------- ------------------------ ---------------------------------- --------------- --------------------
----
          1 Priya                    priya@gmail.com                    9876543210      Chennai
```

4. Apply Filtering Conditions

Multiple conditions can be combined using AND, OR, and NOT.

4.1 AND Condition

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE
  2  FROM PRODUCT
  3  WHERE PRICE > 100
  4  AND PRICE < 1000;

PRODUCT_ID PRODUCT_NAME                      PRICE
---------- ------------------------------ ----------
         1 Gold Plated Necklace              599.00
         2 Diamond Earrings                  799.00
         3 Pearl Bracelet                    450.00
         4 Silver Anklet                     350.00
         5 Stone Ring                        299.00
```

4.2 OR Condition

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE
  2  FROM PRODUCT
  3  WHERE PRICE < 500
  4  OR PRICE > 1000;

PRODUCT_ID PRODUCT_NAME                    PRICE
---------- ------------------------------ ----------
         3 Pearl Bracelet                 450.00
         4 Silver Anklet                  350.00
         5 Stone Ring                     299.00
```

4.3 IN Operator

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE
  2  FROM PRODUCT
  3  WHERE PRODUCT_ID IN (1, 2, 3);

PRODUCT_ID PRODUCT_NAME                    PRICE
---------- ------------------------------ ----------
         1 Gold Plated Necklace           599.00
         2 Diamond Earrings               799.00
         3 Pearl Bracelet                 450.00
```

4.4 BETWEEN Operator

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE
  2  FROM PRODUCT
  3  WHERE PRICE BETWEEN 500 AND 1500;

PRODUCT_ID PRODUCT_NAME                    PRICE
---------- ------------------------------ ----------
         1 Gold Plated Necklace           599.00
         2 Diamond Earrings               799.00
```

5. Basic Business Reports

5.1 Product Price Report

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE
  2  FROM PRODUCT
  3  ORDER BY PRICE DESC;

PRODUCT_ID PRODUCT_NAME                    PRICE
---------- ------------------------------ ----------
         2 Diamond Earrings                799.00
         1 Gold Plated Necklace            599.00
         3 Pearl Bracelet                  450.00
         4 Silver Anklet                   350.00
         5 Stone Ring                      299.00
```

5.2 Customer Order Report

```
SQL> SELECT
  2      CUSTOMER_ID,
  3      ORDER_ID,
  4      ORDER_DATE,
  5      TOTAL_AMOUNT,
  6      ORDER_STATUS
  7  FROM ORDERS
  8  ORDER BY ORDER_ID;

CUSTOMER_ID   ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
----------- ---------- --------- ------------ ------------------------------
          1       1001 22-SEP-26         1198 Shipped
          2       1002 22-SEP-26          999 Packed
          3       1003 23-SEP-26          599 Shipped
          4       1004 23-SEP-26          449 Delivered
          5       1005 24-SEP-26          349 Pending
          1       1006 25-SEP-26          799 Pending

6 rows selected.
```

5.3 Product Order Report

```
SQL> SELECT
  2      CUSTOMER_ID,
  3      COUNT(*) AS TOTAL_ORDERS
  4  FROM ORDERS
  5  GROUP BY CUSTOMER_ID
  6  ORDER BY TOTAL_ORDERS DESC;

CUSTOMER_ID TOTAL_ORDERS
----------- ------------
          1            2
          5            1
          4            1
          2            1
          3            1
```

5.4 Customer-wise Total Orders

```
SQL> SELECT
  2      CUSTOMER_ID,
  3      SUM(TOTAL_AMOUNT) AS TOTAL_PURCHASE_AMOUNT
  4  FROM ORDERS
  5  GROUP BY CUSTOMER_ID
  6  ORDER BY TOTAL_PURCHASE_AMOUNT DESC;

CUSTOMER_ID TOTAL_PURCHASE_AMOUNT
----------- --------------------
          1                 1997
          2                  999
          3                  599
          4                  449
          5                  349
```